4 Fig. 4.1 shows a stainless-steel saucepan being heated on an electric cooker. The saucepan contains water.

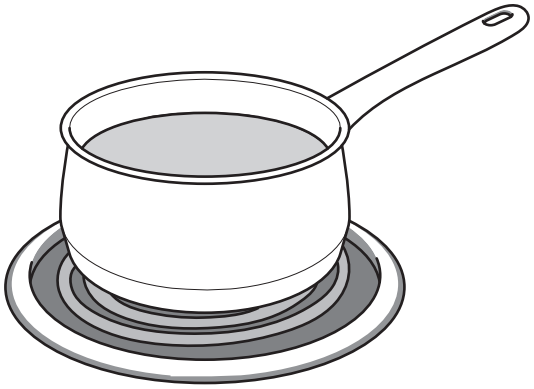


Fig. 4.1

(a) State what happens to the water particles as the water temperature increases.

..... [1]

(b) The saucepan contains 250 cm^3 of water. The specific heat capacity of water is 4200 J / (kg °C). The density of water is 1000 kg / m^3.

(i) Show that the mass of the water in the saucepan is 0.25 kg.

[2]

(ii) Calculate the energy required to increase the water temperature from 20 °C to 65 °C.

energy = ... [3]

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(iii) The heater supplies enough power to heat the water in 39 s.
A student measures the time taken to heat the water as 115 s.

Suggest why the actual time taken to heat the water is longer. Assume that the student takes accurate measurements.

.....

..... [1]

(c) The stainless-steel saucepan is replaced with an aluminium saucepan of the same mass. It contains the same volume of water.

The specific heat capacity of stainless steel is 500 J / (kg °C).
The specific heat capacity of aluminium is 890 J / (kg °C).

Explain how using an aluminium saucepan will affect the time taken to heat the water.

.....

.....

..... [2]

[Total: 9]